

hw 3, env s 193ds

Damian Rojas

```
#read packages
library(tidyverse)
library(janitor)
library(here)

#store kelp data
kelp <- read.csv("data/temp-kelp.csv")

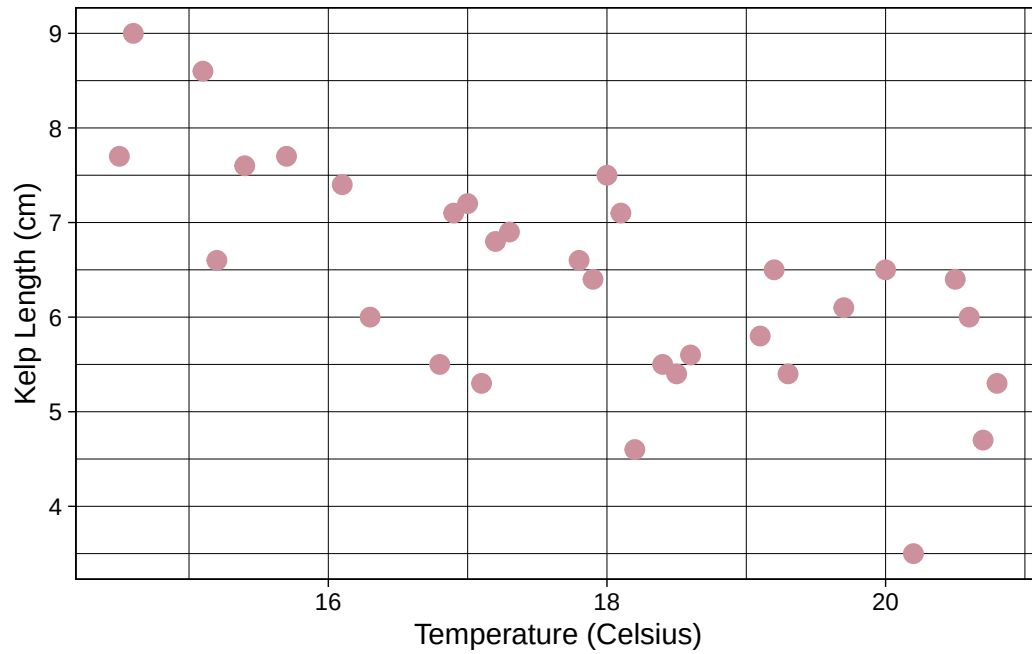
#store personal data
walks <- read.csv("data/walks_data.csv")
```

1)

a)

The appropriate tests we can run to test the strength of the relationship between temperature and giant kelp frond elongation rate are a Pearson's correlation test and a Spearman's rank correlation. The differences between the two tests are that a Pearson's test observes the strength between linear relationships while the Spearman's test observes the strength between monotonic relationships that do not need to be linear.

b) Visualizations



c) Checking Assumptions / Run test

```
#create linear model for kelp data
kelp_model <- lm (kelp_elong ~ temp_c, data = kelp)

#show lm for kelp
kelp_model
```

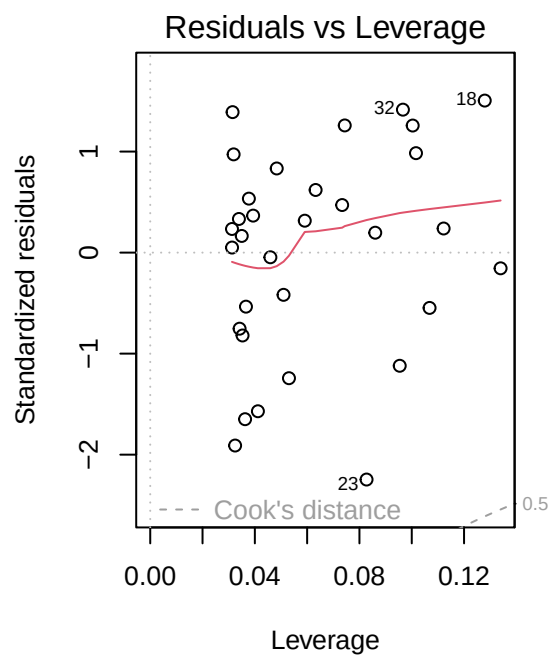
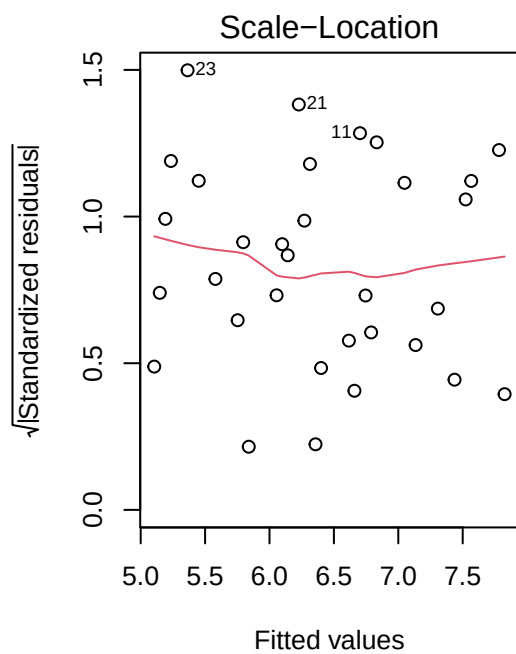
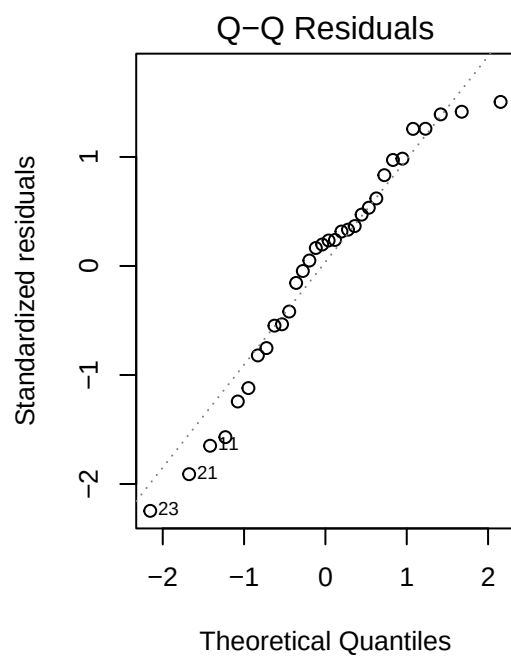
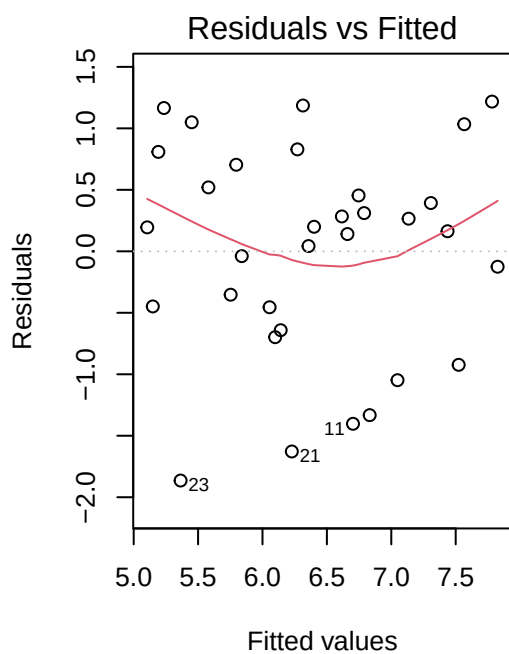
Call:

```
lm(formula = kelp_elong ~ temp_c, data = kelp)
```

Coefficients:

(Intercept)	temp_c
14.0864	-0.4318

```
#kelp elongation as function of temp
#
#plotted as 2x2 grid
par(mfrow = c(2,2))
#summary plots of kelp model
plot (kelp_model)
```



Checked normality, homoscedasticity, and any outliers of data with summary plots. Found to be normal, linear, with no outliers affecting data.

```
#pearsons cor test is the method for linear models
#run test with x as kelp elongation, y as temp, method as pearson
cor.test(x = kelp$kelp_elong, y = kelp$temp_c, method = "pearson")
```

Pearson's product-moment correlation

```
data: kelp$kelp_elong and kelp$temp_c
t = -5.189, df = 30, p-value = 1.366e-05
alternative hypothesis: true correlation is not equal to 0
95 percent confidence interval:
 -0.8359665 -0.4460157
sample estimates:
      cor
-0.6877489
```

d)

To evaluate the strength of the relationship between temperature and giant kelp frond elongation rate, I used a Pearson's correlation test. We found a strong negative relationship between kelp elongation and temperature (Pearson's $r = -0.69$, $t(30) = -5.19$, $p < 0.001$), where temperature significantly predicted kelp elongation length ($R^2 = 0.476$)